

Lukas Lamster

PhD Student · ISEC · Graz University of Technology

✉ lukas.lamster@tugraz.at ☎ +43 664 352 8681 📍 Graz, Austria
🌐 lamsterl 🆔 0000-0003-4046-8727 🎓 Google Scholar 🏛 Institute profile

Research Interests

I am a security researcher focusing on *improving runtime security through novel hardware-based mechanisms*. One of my main research topics is the combination of memory integrity protection with tagged memory architectures. I explore how synergies between these technologies can be used to provide more efficient and stronger tagging, while still allowing for error detection and correction.

Education

Mar 2022 – present	PhD in Computer Science <i>Graz University of Technology</i> Advisor: Prof. Stefan Mangard Expected: Spring 2027	Graz, Austria
2019 – 2021	MSc in Computer Science <i>Graz University of Technology</i> Thesis: <i>DRAM Integrity Protection with Efficient Low-Latency Cryptographic Primitives</i> Received the Student Research Excellence Award Received the WKO Research Stipend Graduated with distinction.	Graz, Austria
2015 – 2019	BSc in Computer Science <i>Graz University of Technology</i> Thesis: <i>From Rowhammer to Nethammer – Utilizing Intel CAT to Remotely Induce Bit Flips</i> Received the Student Research Excellence Award Graduated with distinction.	Graz, Austria

Research Experience

Mar 2022 – present	Doctoral Researcher <i>Institute of Information Security, Graz University of Technology</i> • Exploring combinations of DRAM integrity protection and memory tagging • Developing novel hardware-based runtime security mechanisms • Multiple contributions towards memory safety, efficient memory tagging, and DRAM integrity protection schemes	Graz, Austria
Summer 2019	Research Intern <i>ISEC, Graz University of Technology</i> • Improving the reverse engineering of processor-specific DRAM mapping functions	Graz, Austria

Work Experience

Mar – Jun 2026	Software Development Intern <i>Cloudflare</i> • Researching Rowhammer on large-scale computing infrastructure • Researching side channels and timers in Worker and Container products	Lisbon, Portugal
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2018 – 2020	Software Engineer AVL • Jira plugin development • Developing compatibility layers for internal tooling	Graz, Austria
2016 – 2017	Software Engineer <i>Melecs EWS</i> • Development of drivers for assembly line equipment • Quality assurance tool development	Siegingdorf, Austria

Selected Publications

USENIX Security 2024	Voodoo: Memory Tagging, Authenticated Encryption, and Error Correction through MAGIC ▶ Voodoo extends the MAGIC mode with memory tagging without additional storage overheads. Lukas Lamster , Martin Unterguggenberger, David Schrammel, Stefan Mangard
IEEE S&P 2023	CSI:Rowhammer – Cryptographic Security and Integrity against Rowhammer ▶ Protecting DRAM data integrity with MACs allows the detection of virtually all bit flips. An iterative correction procedure enables error recovery. Jonas Juffinger, Lukas Lamster , Andreas Kogler, Maria Eichlseder, Moritz Lipp, Daniel Gruss
USENIX Security 2023	HashTag: Hash-based Integrity Protection for Tagged Architectures ▶ Replacing linear codes with hashes allows us to implement tagged memory without additional fetches or a carve-out region. Lukas Lamster , Martin Unterguggenberger, David Schrammel, Stefan Mangard

Talks & Presentations

Sep 2025	WaitWatcher & WaitGuard: Detecting Flush-Based Cache Side-Channels Through Spurious Wakeups. Paper presentation, 30th European Symposium on Research in Computer Security (ESORICS 2025), Toulouse, France.
Aug 2025	FatPTE: Expanding Page Table Entries for Security. Paper presentation, 20th International Conference on Availability, Reliability and Security (ARES 2025), Ghent, Belgium.
Aug 2024	Voodoo: Memory Tagging, Authenticated Encryption, and Error Correction through MAGIC. Paper presentation, 33rd USENIX Security Symposium, Philadelphia, PA, USA.
Aug 2023	HashTag: Hash-based Integrity Protection for Tagged Architectures. Paper presentation, 32nd USENIX Security Symposium, Anaheim, CA, USA.

Skills

Programming	C, C++, C#, Python, x86-64 assembly, RISC-V assembly, SystemVerilog
Tools	git, gdb, pwndbg, Docker, LLM-assisted development, Ghidra
Projects	Linux kernel development, gem5 development, qemu, Spike RISC-V ISA Simulator, OpenSBI
Languages	German (native), English (fluent)

Teaching

2025 – present	Secure System Architecture, Lecture and Practicals <i>Graz University of Technology</i>
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Lectures, practical sessions, and grading for approx. 40 students per year. We cover topics ranging from memory tagging and pointer authentication to trusted computing and confidential virtual machines.

2022 – present

Information Security, Practicals

Graz University of Technology

Introductory information security course, teaching approx. 400 students per year. My focus is on the system security part, which consists of CTF-style binary exploitation tasks.

2022 – present

Computer Organization and Networks, Practicals

Graz University of Technology

Introductory computer organization course for Bachelor's students, teaching approx. 500 students per year. Course contents cover hardware design in SystemVerilog, designing and implementing state machines, integrating extensions into existing hardware designs, and RISC-V assembly.

Awards

2025

Distinguished Artifact Reviewer Award, USENIX Security 2025.

2021

WKO Research Stipend, Master's Thesis, WKO.

2021

Student Research Excellence Award, Master's Thesis, ISEC.

2019

Student Research Excellence Award, Bachelor's Thesis, ISEC.